

# M13-C3 — H6 folding-steps and nested- $n$ ablation (search cost under `baseline_cautious`)

H6 run / analysed / not a Bucket 3 claim

## 1 Status

**H6 run / analysed.** Bounded procedural investigation of **search cost** under repository `baseline_cautious` (`cautious + nd / any / all / relto`) on fixture `m13_bucket3_binary_collider_and`. Markdown twin: `h6_folding_and_n_ablation.md`.

**H6a lead:** at fixed `n30_seed42`, root ( $a/b$ ) trace length is approximately linearly proportional to permitted `folding_steps( $M$ )`; extra budget beyond the failed motif is wasted search time.

**H6b lead:** at fixed `folding_steps(2)`, root trace length is approximately linearly proportional to sample size  $n$  (topology fixed; per-band bookkeeping grows). Roots are procedural diagnostics (evaluator `no_observed_parent_deterministic_rule`); only child  $c$  is a desirable deterministic mechanism target. H0–H5 closed / unchanged. **Not a Bucket 3 claim.** H7a analysed (`h7a_relto_vs_sechk.md`); H7b not started.

## 2 Research question

Under repository-baseline `cautious + nd` on the closed H0 AND-collider table, how do permitted `folding_steps( $M$ )` (H6a) and nested sample size  $n$  at fixed  $M=2$  (H6b) drive search cost for roots without recoverable deterministic parent rules, and what happens to control  $c$ ?

## 3 Main findings

### 3.1 H6a — folding-steps ablation

On frozen `n30_seed42`, for roots  $a/b$ , `prolog.stdout` length grows roughly linearly with  $M \in \{1, 2, 5, 10\}$  ( $\sim 549 \rightarrow 1047 \rightarrow 2541 \rightarrow 5031$  lines for  $a$ ; mirrored for  $b$ ). Outcome stays `completed_no_solution` at every  $M$ . Control  $c$  solves inside `tokens(1)` with identical delta and unchanged 134-line /  $\sim 1.3$ s trace at all four budgets.

**Why predictable.** `folding_steps( $M$ )` does *not* mean “perform exactly  $M$  folds.” Under `nd`, learning starts at `tokens(1)`. When a full attempt under the current token budget fails, `gen.pl` may print `* Increasing folding tokens to: T1` and retry while  $T_1 \leq M$ . On these roots, each budget replays essentially the same failed cautious motif ( $\sim 9$  NEW assumptions /  $\sim 500$ -line band). Higher  $M$  multiplies failed-band replays: lines  $\approx C_{\text{fixed}} + M \cdot C_{\text{band}}$ . There is no successful  $M^*$  through  $M=10$  — raising  $M$  does not buy a new theory; it only replays failure (H4 wasted headroom).

### 3.2 H6b — nested sample-size ablation

At fixed `folding_steps(2)`, for roots  $a/b$ , trace length grows roughly with  $n \in \{30, 60, 90\}$  ( $\sim 1047 \rightarrow 1641 \rightarrow 2241$  lines for  $a$ ). Search topology is fixed ( $\text{NEW}=18$ ,  $\text{KO}=32$ ; one token increase to 2). What grows is per-band bookkeeping (more grounded  $E^\pm$  ids; larger rote/`ert` and subsumption sweeps; larger contrary batches).

**Why predictable.** With motif  $\times$  budget structure held fixed, cost per band should scale roughly with grounding / example size  $\Rightarrow$  total length roughly  $\propto n$  in this limited range. Support *presence* of the four atoms is already complete at  $n=30$ ; multiplicities and computational grounding size change. H6b is not “pure compute with every statistical property fixed.” Line ratios are near-linear; runtime slightly more superlinear. Do not claim a universal complexity law.

### 3.3 Cross lead

H4’s long root traces ( $\sim 5\text{k}$  lines /  $\sim 66\text{--}70\text{s}$  at  $n30 \times \text{steps}10$ ) are primarily cumulative nd token-budget replays of one cautiously failed motif (H6a), with a secondary per-band cost that grows with  $n$  at fixed  $M=2$  (H6b); control  $c$  stays solved and delta-stable; nothing here is root mechanism recovery or an  $n \times \text{steps}$  interaction result.

## 4 Setup

Fixture `m13_bucket3_binary_collider_and`. Bundle: `cautious + nd / any / all / relto`. Base Prolog `configs/baseline_cautious_config.pl` (`steps10`; unchanged); ablation Prolog `configs/baseline_cautious_steps{1,2,5}_config.pl`. H6a: fixed `n30_seed42`,  $M \in \{1, 2, 5, 10\}$  (`steps10` = reused H4). H6b: fixed  $M=2$ ; nested `n30/n60/n90` seed 42. `n30 sha256 1edae465d0f6b6b662b3c4ed4863af21c1d9b0c1cf428774731dd3f43282a8ca`; `n30  $\subset$  n60  $\subset$  n90`. Atom counts  $(0, 0, 0)/(0, 1, 0)/(1, 0, 0)/(1, 1, 1)$ : `n30 2/4/7/17`; `n60 4/10/12/34`; `n90 5/17/18/50`.

Paths: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/<config>/<sample>`  
H4 reuse summary `sha256 1e4b4199fe03a5a6208967b092450f67717d4e0dc788271f771c0d18b1ff23de`.

Representative commands:

```
python -m causal.targetwise.cli run \
  --config ../baseline_cautious_steps1/n30_seed42.yaml
python -m causal.targetwise.cli run \
  --config ../baseline_cautious_steps2/n{30,60,90}_seed42.yaml
python -m causal.targetwise.cli run \
  --config ../baseline_cautious_steps5/n30_seed42.yaml
```

Engine gate (read-only): `gen.pl` increases `tokens(T)` while  $T+1 \leq M$  under `folding_mode(nd)`; initialisation `tokens(1)`.

## 5 Verified outcome table

Recorder-verified against `metrics.json` (`aba_learning_runtime_s`) and `output/prolog.stdout`.

| sample | $M$ | $t$ | outcome | rt(s)  | lines | incs     | NEW | KO  |
|--------|-----|-----|---------|--------|-------|----------|-----|-----|
| n30    | 1   | $a$ | no-sol  | 6.638  | 549   | none     | 9   | 16  |
| n30    | 1   | $b$ | no-sol  | 6.797  | 555   | none     | 9   | 16  |
| n30    | 1   | $c$ | solved  | 1.304  | 134   | —        | 1   | 1   |
| n30    | 2   | $a$ | no-sol  | 13.117 | 1047  | to 2     | 18  | 32  |
| n30    | 2   | $b$ | no-sol  | 13.487 | 1062  | to 2     | 18  | 32  |
| n30    | 2   | $c$ | solved  | 1.279  | 134   | —        | 1   | 1   |
| n30    | 5   | $a$ | no-sol  | 32.775 | 2541  | to 2..5  | 45  | 80  |
| n30    | 5   | $b$ | no-sol  | 32.825 | 2583  | to 2..5  | 45  | 80  |
| n30    | 5   | $c$ | solved  | 1.271  | 134   | —        | 1   | 1   |
| n30    | 10  | $a$ | no-sol  | 69.952 | 5031  | to 2..10 | 90  | 160 |
| n30    | 10  | $b$ | no-sol  | 66.473 | 5118  | to 2..10 | 90  | 160 |
| n30    | 10  | $c$ | solved  | 1.288  | 134   | —        | 1   | 1   |
| n60    | 2   | $a$ | no-sol  | 22.169 | 1641  | to 2     | 18  | 32  |
| n60    | 2   | $b$ | no-sol  | 22.617 | 1659  | to 2     | 18  | 32  |
| n60    | 2   | $c$ | solved  | 2.126  | 200   | —        | 1   | 1   |
| n90    | 2   | $a$ | no-sol  | 32.291 | 2241  | to 2     | 18  | 32  |
| n90    | 2   | $b$ | no-sol  | 32.894 | 2262  | to 2     | 18  | 32  |
| n90    | 2   | $c$ | solved  | 3.033  | 266   | —        | 1   | 1   |

Integrity: H6a n30 BK/examples/data identical across steps 1/2/5/10 per target; control  $c$  delta.abi byte-identical on all six collections (sha256 f9f75b60631b2a2a...426865a):

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

## 6 Analysis A (H6a)

Decisive cautious KO already inside budget-1 band:

```
folding result: c_alpha_3(A) <- [b_val_0(A)]
found: alpha_2/1 ... KO, cannot introduce an assumption for [b_val_0(A)]
```

NEW=9  $\times$   $M$ . Steps1 body after the lopt(...) dump equals steps2 band-0 for 522 lines; then steps1 prints *\* No solution found!* while steps2 prints *\* Increasing folding tokens to:* 2. Rough linearity for  $a$ : lines/ $M \approx 549, 524, 508, 503$  across  $M=1, 2, 5, 10$ .

Control  $c$ : solved every  $M$ ; 134 lines /  $\sim 1.3$ s; no token increases.

## 7 Analysis B (H6b)

Fixed  $M=2$ . Topology fixed (NEW=18, KO=32).  $a$  lines 1047 $\rightarrow$ 1641 $\rightarrow$ 2241; runtime 13.1 $\rightarrow$ 22.2 $\rightarrow$ 32.3s.  $c$  same delta bytes; lines 134 $\rightarrow$ 200 $\rightarrow$ 266; runtime 1.28 $\rightarrow$ 2.13 $\rightarrow$ 3.03s. Support presence complete at  $n=30$ ; multiplicities change.

## 8 Cross-analysis vs H4 / H5

H4's  $\sim 5$ k-line /  $\sim 70$ s root no-sols are the  $M=10$  endpoint of H6a (ten failed-band replays). Secondary H5 pointer: **greedy\_cautious** H0  $a$  reaches the same no-sol in  $\approx 344$  lines / 0.42s via a different search bundle — H4 cost is not “cautious alone.”

## 9 Boundaries / non-claims

- Not a Bucket 3 claim.
- Not “cautious is better.”
- Root no-solution is not mechanism recovery.
- Budget-relative no-solution  $\neq$  unlearnability at larger  $M$ .
- Control  $c$  solved / delta match is not automatic causal recovery.
- No  $n \times \text{folding\_steps}$  interaction claim (grid not run).
- Approximate linearity only in these limited slices.
- H7a analysed; H7b not started; H0–H5 readings preserved.

## 10 Next

H6 documentation complete. H7a analysed (`h7a_relto_vs_sechk.md`); H7b not started. Still **no Bucket 3 claim**.